

PERFORMANCE WORK STATEMENT

Rockwell Automation Allen Bradley PLC and HMI Training, CCP 146, CCP 151, CCP 143, CCV 204-A, T-CLXPT

PART 1 GENERAL INFORMATION

1. **GENERAL:** This is a non-personal services contract to provide Rockwell Automation Allen Bradley Programmable Logic Controller (PLC), Human Machine Interface (HMI), and ControlLogix training. The Government shall not exercise direct supervision or control over any Contractor employees. Notwithstanding Contractor shall ensure employees comply with all requirements of this Performance Work Statement (PWS) and contract.

1.1 Description of Services/Introduction: The Contractor shall provide all personnel, equipment, supplies, instruction, transportation, tools, materials, supervision, other items and non-personal services necessary to perform Rockwell Automation Allen Bradley PLC, HMI, and ControlLogix Programmer training. The instruction shall include the Rockwell Automation Allen Bradley classes CCP-146, CCP-151, CCP-143, CCV204-A and T-CLXPT as defined in this Performance Work Statement except for those items specified as government furnished property and services. The Contractor shall perform to the standards in this contract.

1.2 Background:

PLCs and HMIs are industrial control devices that are used in a variety of electromechanical systems. Examples of such systems include use in the pump valve test facility, [REDACTED] sensor management, and for industrial process control.

All current and near-term PLC equipment purchased and/or will be purchased by Code [REDACTED] are Rockwell Automation Allen Bradley brand. Proficiency with this equipment is crucial. Code [REDACTED] engineers and technicians must learn how to operate and program it. This training will assist with better project outcomes, faster, and more adaptable service and much more cost-effective solutions. This course of instruction directly supports the mission of Puget Sound Naval Shipyard and Intermediate Maintenance Facility (PSNS&IMF).

1.3 Objectives:

The objective is for Code [REDACTED] engineers and technicians to learn how to program Rockwell Automation Allen Bradley brand PLCs and HMIs using Rockwell Automation training. PSNS&IMF uses exclusively the Rockwell Automation Allen Bradley brand. For this effort we will be using the Rockwell Automation Instructor to deliver hands-on training at PSNS&IMF. The courses will be:

- CCP-146 Studio 5000 Logix Designer Level 1: Control Logix System Fundamentals—this training will help personnel develop and build a solid foundation with a fundamental knowledge of ControlLogix and other Logix5000 systems. Personnel will be introduced to basic Logix5000 concepts and terminology. Students will be exposed to Logix5000 system hardware, including hands-on experience with the ControlLogix platform. Students will use a simulated workstation using a combination of Studio 5000 Logix Emulate software and Factory Talk View SE software to provide the I/O simulator, and/or Factory Talk Linx or RSLinx Classic software to provide the communication functions.
- CCP-151 Studio 5000 Logix Designer Level 2: Basic Ladder Logic Programming—this is a skill building course that provides personnel with resources and hands-on practice. The student will

use the Studio 5000 Logix Designer application to perform basic software tasks to meet the requirements of a given functional specification. The student will set up a sequencer to run equipment through a predefined procedure and separate production procedure from equipment control.

This course will use a simulated workstation using a combination of Studio 5000 Logix Emulate software and FactoryTalk View SE software to provide I/O simulator, and/or Factory Talk Linx or RSLinx Classic software to provide the communication functions.

- CCP-143 Studio 5000 Logix Designer Level 3: Project Development—when given a functional specification for a Logix5000 application, the student should be able to develop a project to meet the specification requirements. This course covers tasks common to the following hardware that uses Logix5000 as a control engine:
 - ControlLogix controllers
 - CompactLogix controllers
 - SoftLogix controllers

This course builds upon the knowledge of common controller terms, operation, and the student's experience with basic ladder logic programming. The student will gain a deeper understanding of project development tasks that are common to all Logix controllers. Such tasks include organizing tasks and routines, organizing controller data, configuring modules, and sharing data. The student will also use producer/consumer technology for multicasting to input and output devices, sharing data between controllers, and controlling remote I/O. This training will use simulated workstations using a combination of Studio5000 Logix Emulate Software, FactoryTalk View SE software to provide the I/O simulator, and/or FactoryTalk Linx or RSLinx Classic software to provide communication functions.

- CCV204-A FactoryTalk ME and PanelView Plus Programming --given a computer with FactoryTalk View Machine Edition (ME) software and a PanelView Plus 7 terminal, the student should be able to produce an operational application running on the terminal. During the class the student will practice the following hands-on skills:
 - Prepare a PanelView Plus 7 terminal for operation
 - Create a new application and run it on a terminal
 - Create, configure, and animate graphic objects on graphic displays
 - Configure security for granting/restricting access to certain graphic displays or for rights to perform certain actions
 - Create and configure messages and alarms for alerting operators to changes in a process.

1.4 Scope: The service provided by the Contractor will be providing the following Rockwell Automation classes: CCP 146, CCP 151, CCP 143, CCV 204-A, and T-CLXPT. These classes shall be provided by a Rockwell Automation authorized distributor and hosted with an instructor at Puget Sound Naval Shipyard and Intermediate Maintenance Facilities (PSNS & IMF).

1.5 Period of Performance: All classes and the assessment shall be completed within 63 days after contract award.

1.6 Quality Assurance: The government shall evaluate the Contractor's performance under this contract in accordance with the Quality Assurance Surveillance Plan. This plan is primarily focused on what the Government must do to ensure that the Contractor has performed in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rate(s).

1.7 Location:

1.7.1 Place of Performance: The work to be performed under this contract will be performed at [REDACTED], Apprentice School, on Naval Base Kitsap.

1.7.2 Hours of Operation: The Contractor is responsible for conducting business, between the hours of 0720 - 1530 Monday thru Friday except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. For other than firm fixed price contracts, the Contractor will not be reimbursed when the government facility is closed for the above reasons. The Contractor must at all times maintain an adequate workforce for the uninterrupted performance of all tasks defined within this PWS when the Government facility is not closed for the above reasons. When hiring personnel, the Contractor shall keep in mind that the stability and continuity of the workforce are essential.

1.7.3 Recognized Holidays: Contractor is not required to perform services on federal holidays. The list of current federal holidays can be found here: <https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/>

In addition to the above listed holidays, Puget Sound Naval Shipyard is normally closed for all except necessary services during the week between Christmas Day and New Year's Day (curtailment). Necessary services/ frequencies for various facilities are noted on the detail sheets and work contracted for shall be performed during this period unless otherwise directed by the Contracting Officer. Exact number of closure days may vary from year to year. Actual requirements will be determined in advance of the closure. On site work performed by the Contractor for the Contractor's convenience during this time must also be specifically requested by the Contractor in writing and approved by the Contracting Officer in advance.

Contractor Employees, in order to gain access to the government facility identified in paragraph 1.7.1 to perform work under this contract, shall comply with the Badging and Access Requirements Attachment to the solicitation/contract.

1.8.1 Physical Security: The Contractor shall be responsible for safeguarding all government equipment, information and property provided for Contractor use. At the close of each work period, government facilities, equipment, and materials shall be secured.

1.8.2 Key Control: The Contractor shall establish and implement methods of making sure all keys/key cards issued to the Contractor by the Government are not lost or misplaced and are not used by unauthorized persons. NOTE: All references to keys include key cards.

No keys issued to the Contractor by the Government shall be duplicated. The Contractor shall develop and implement procedures covering key control to meet requirements of this section. Such procedures shall include turn-in of any issued keys by personnel who no longer require access to locked areas. The Contractor shall immediately report any occurrences of lost or duplicate keys/key cards to the Contracting Officer.

In the event keys, are lost or duplicated, the Contractor shall, upon direction of the Contracting Officer, re-key or replace all affected locks; however, the Government, at its option, may replace the affected lock or locks or perform re-keying. When the replacement of locks or re-keying is performed by the Government, the total cost of re-keying or the replacement of the lock or locks shall be deducted from payment due the Contractor.

The Contractor shall prohibit the use of Government issued keys by any persons other than the Contractor's employees. The Contractor shall prohibit the opening of locked areas by Contractor employees to permit entrance of persons other than Contractor employees engaged in the performance of work required by this PWS defined herein.

1.9 Special Qualifications: The Contractor providing the instructor and class services shall be a Rockwell Automation authorized distributor per Rockwell Automation website:
<https://www.rockwellautomation.com/en-us/sales/partner-locator.html>.

1.10 Post Award Conference/Periodic Progress Meetings: The Contractor agrees to attend any post award conference convened by the contracting activity or contract administration office in accordance with FAR Subpart 42.5. The contracting officer, Contracting Officers Representative (COR), and other Government personnel, as appropriate, may meet periodically with the Contractor to review the Contractor's performance. Appropriate action shall be taken to resolve outstanding issues. These meetings shall be at no additional cost to the government.

1.11 Contracting Officer Representative (COR): The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following functions: assure that the Contractor performs the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, specifications; monitor Contractor's performance and notifies both the Contracting Officer and Contractor of any deficiencies; coordinate availability of government furnished property, and provide site entry of Contractor personnel.

The COR is NOT authorized to change any of the terms and conditions of this contract. **Only the warranted Contracting Officer is authorized to make modifications/changes to the terms and conditions of this contract.** If the Contractor believes any Government personnel has effectively issued a change resulting in a change in effort and price/cost of the contract, the Contractor shall immediately notify the Contracting Officer in writing.

CORS:

PRIMARY

[REDACTED]
[REDACTED]
[REDACTED]

Government Technical Points of Contact (TPOC): The TPOC(s) monitor all technical aspects of the contract and assists in contract administration. The TPOC is authorized to perform the following functions: assure that the Contractor performs the technical requirements of the contract; perform inspections necessary in connection with contract performance; maintain written and oral communications with the Contractor concerning technical aspects of the contract; issue written interpretations of technical requirements, including Government drawings, designs, specifications; monitor Contractor's performance and notify both the Contracting Officer and Contractor of any deficiencies; coordinate availability of Government furnished property; and provide site entry of Contractor personnel.

The TPOC(s) are NOT authorized to change any of the terms and conditions of this contract. **Only the warranted Contracting Officer is authorized to make modifications/changes to the terms and conditions of this contract.** If the Contractor believes any Government personnel has effectively issued a change resulting in a change in effort and price/cost of the contract, the Contractor shall immediately notify the Contracting Officer in writing.

TPOCS:

PRIMARY

[REDACTED]
[REDACTED]
[REDACTED]

ALTERNATE

[REDACTED]
[REDACTED]
[REDACTED]

1.12 Contract Manager: The Contractor shall provide a contract manager who shall be responsible for the performance of the work. The name of this person and an alternate who shall act for the Contractor when the manager is absent shall be designated in writing to the contracting officer. The contract manager or alternate shall have full authority to act for the Contractor on all contract matters relating to execution of this contract.

1.12.1 Identification of Contractor Employees: All Contractor personnel working in situations where their Contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression that they are Government officials. All Contractor personnel on a government installation will be required to obtain and wear badges in the performance of this contract.

PART 2 DEFINITIONS & ACRONYMS

2. DEFINITIONS AND ACRONYMS:

2.1. COMMON DEFINITIONS: Per Federal Acquisition Regulation (FAR) Clause 52.212-4(e), clause 52.202-1 is incorporated by reference to this contract.

2.2 SPECIAL DEFINITIONS:

2.2.1 Programmable Logic Controller: An industrial computer used to automate and control electromechanical processes.

2.2.2 Human Machine Interface: The hardware and software used to facilitate communication between human operators and PLCs.

2.2.3 Studio 5000 Logix Designer: Rockwell Automation's design and configuration software for Rockwell Automation's Logix PLCs.

2.2.4 ControlLogix: A product line series of PLCs in the Logix product family of PLCs.

2.2.5 Ladder Logic: A graphical programming language used to program PLCs.

2.2.6 Tags: Metadata assigned to data to make it identifiable and categorizable.

2.2.7 Ethernet/IP: An industrial communications protocol.

2.2.8 Array: A data structure that stores multiple values of the same data type.

2.2.9 FactoryTalk View ME: Rockwell Automation's software for building and running Rockwell Automation HMIs.

2.2.10 PanelView Plus: A Rockwell Automation HMI product line.

2.3 Acronyms:

PLC	Programmable Logic Controller
PWS	Performance Work Statement
HMI	Human Machine Interface
TPOC	Technical Point of Contact
FAR	Federal Acquisition Regulation
QASP	Quality Assurance Surveillance Plan
PSNS	Puget Sound Naval Shipyard

PSNS&IMF	Puget Sound Naval Shipyard and Intermediate Maintenance Facilities
CIA	Controlled Industrial Area
NBK	Naval Base Kitsap
I/O	Input/Output
GSV/SSV	Get System Value/Set System Value
BTD	Bit Field Distribute
ME	Machine Edition

PART 3 GOVERNMENT FURNISHED PROPERTY, EQUIPMENT, AND SERVICES

3. GOVERNMENT FURNISHED ITEMS AND SERVICES:

3.1 Facilities: The Government will provide a minimum 700 square foot classroom outside the Puget Sound Naval Shipyard Controlled Industrial Area, desk space for each student, and seating for 12 students.

3.2 Utilities: The Government will provide wired or wireless internet access available to all Rockwell Automation computers provided during the training and 110V power outlets for 6 workstations. The Contractor shall instruct employees in utilities conservation practices. The Contractor shall be responsible for operating under conditions that preclude the waste of utilities, which include turning off the water faucets or valves after using the required amount to accomplish cleaning vehicles and equipment.

3.3 Equipment: The Government will provide two (2) whiteboards with markers, a projector, and a projector screen.

PART 4 CONTRACTOR FURNISHED ITEMS AND SERVICES

4. CONTRACTOR FURNISHED ITEMS AND RESPONSIBILITIES:

4.1 General: The Contractor shall furnish all supplies, equipment, facilities and services required to perform work under this contract that are not listed under Section 3 of this PWS.

4.2. Materials: The Contractor shall furnish training materials for course delivery, 6 training workstations for student hands-on practice, application software for all workstations, and certificates of completion for each student.

PART 5 SPECIFIC TASKS

5. SPECIFIC TASKS:

5.1. Basic Services: The Contractor shall provide setup, configuration, teardown, and shipping of all training support equipment. The Contractor shall provide the instructor and provide the instructor with their own living expenses.

5.2. Session 1: The Contractor shall deliver on-site training for 12 students at Puget Sound Naval Shipyard for the following classes: CCP 146 and CCP 151.

5.2.1 CCP 146 – Studio 5000 Logix Designer Level 1: ControlLogix System Fundamentals. Class days: 2. Student manual and job aid: Digital.
Course learning objectives:

- Understand Control Systems
- Locate ControlLogix System Components
- Locate and Configure Studio 5000 Logix Designer Application components
- Create and Modify a Studio 5000 Logix Designer Project
- Select and Connect to Industrial Networks in a Logix5000 System
- Download and Go Online to a Logix5000 Controller
- Locate Input/Output (I/O) Tags and Devices in a ControlLogix System
- Configure Local 1756-I/O Modules in a Studio 5000 Logix Designer Project
- Create Tags and Monitoring Data in a Studio 5000 Logix Designer Project
- Draft Basic Ladder Logic for a Studio 5000 Logix Designer Routine
- Select Basic Ladder Logic Instructions for a Studio 5000 Logix Designer Routine
- Enter Ladder Logic Components in a Studio 5000 Logix Designer Routine
- Integrated Practice - Create and Verify a Studio 5000 Logix Designer Project

5.2.2 CCP 151 – Studio 5000 Logix Designer Level 2: basic Ladder Logic Programming. Class days: 2. Student manual and job aid: Digital.

Course learning objectives:

- Start and test a ladder diagram
- Program timer and counter instructions
- Program compare, move, and math instructions
- Document and search and handle expressions
- Program and separate the procedure from equipment control
- Copy and fill an array

5.3 Session 2: The Contractor shall deliver on-site training for 12 students at Puget Sound Naval Shipyard for the following class: CCP 143.

5.3.1 CCP 143 – Development and Programming of ControlLogix Projects. Class days: 4. Student manual and job aid: Digital.

Course learning objectives:

- Create and organize a project
- Create periodic and event tasks
- Develop an add-on instruction in ladder diagram
- Organize arrays
- Create a user-defined data type
- Import components
- Enter, edit, and verify ladder logic
- Configure a controller to produce and consume data
- Configure controllers to share data over EtherNet/IP
- Communicate with a local 1756-I/O module and over an EtherNet/IP network
- Configure a message
- Allocate connections
- Retrieve and set controller status values with Get System Value/Set System Value (GSV/SSV) instructions
- Program a BTI instruction
- Handle a major fault
- Manage project files
- Update firmware

5.4 Session 3: The Contractor shall deliver on-site training for 12 students at Puget Sound Naval Shipyard for the following class: CCV204-A and T-CLXPT.

5.4.1 CCV 204-A – FactoryTalk View ME and PanelView Plus Programming. Class days: 4. Student manual and job aid: Digital.

Course learning objectives:

- Prepare a PanelView Plus 7 terminal for operation
- Create a new application and run it on a terminal
- Create, configure, and animate graphic objects on graphic displays
- Configure security for granting/restricting access to certain graphic displays or for rights to perform certain actions
- Create and configure messages and alarms for alerting operators to changes in a process

5.4.2 T-CLXPT – ControlLogix Programmer Assessment. Class days: 1. Student manual and job aid: Digital.

PART 6 APPLICABLE PUBLICATIONS

6. APPLICABLE PUBLICATIONS (CURRENT EDITIONS):

6.1 General: The Contractor must abide by all applicable regulations, publications, manuals, and local policies and procedures.

Puget Sound Naval Shipyard & Intermediate Maintenance Facility (PSNS&IMF)

6.2 Contractor Guide to Environmental and Safety Compliance

6.3 OSHAS 18001

PART 7 ATTACHMENT/TECHNICAL EXHIBIT LISTING

7. ATTACHMENT/TECHNICAL EXHIBIT LIST:

7.1 Technical Exhibit 1 – Performance Requirements Summary:

7.2. Technical Exhibit 2 – Deliverables Schedule:

TECHNICAL EXHIBIT 1

Performance Requirements Summary

The Contractor service requirements are summarized into performance objectives that relate directly to mission essential items. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement. These thresholds are critical to mission success.

Performance Objective	Standard	Performance Threshold	Method of Surveillance
PRS # 1. The Contractor shall provide instruction on the learning objectives in CCP 146 listed in paragraph 5.2.1 in this PWS.	The Contractor provided instruction per the learning objectives in CCP 146 listed in paragraph 5.2.1 in this PWS.	Zero deviation from standard.	<i>100 percent inspection.</i>
PRS # 2. The Contractor shall provide instruction on the learning objectives in CCP 151 listed in paragraph 5.2.2 in this PWS.	The Contractor provided instruction per the learning objectives in CCP 151 listed in paragraph 5.2.2 in this PWS.	Zero deviation from standard.	<i>100 percent inspection.</i>
PRS # 3. The Contractor shall provide instruction on the learning objectives in CCP 143 listed in paragraph 5.3.1 in this PWS.	The Contractor provided instruction per the learning objectives in CCP 143 listed in paragraph 5.3.1 in this PWS.	Zero deviation from standard.	<i>100 percent inspection.</i>
PRS # 4. The Contractor shall provide instruction on the learning objectives in CCV 204-A listed in paragraph 5.4.1 in this PWS.	The Contractor provided instruction per the learning objectives in CCV 204-A listed in paragraph 5.4.1 in this PWS.	Zero deviation from standard.	<i>100 percent inspection.</i>
PRS # 5. The Contractor shall provide the ControlLogix Programmer Assessment (T-CLXPT) as stated in paragraph 5.4.2 in this PWS. The Contractor shall provide certificates of completion upon students passing the assessment.	The Contractor provided the ControlLogix Programmer Assessment (T-CLXPT) as stated in paragraph 5.4.2 in this PWS. The Contractor provided certificates of completion upon students' successful completion of the assessment.	Zero deviation from standard.	<i>100 percent inspection.</i>

TECHNICAL EXHIBIT 2

Deliverables Schedule

Deliverable	Frequency	# of Copies	Medium/Format	Submit To
Certificate of completion of training following each student's successful completion of the ControlLogix Programmer Assessment as listed in paragraph 5.4.2 of this PWS	Within ten (10) business days after the successful completion of the ControlLogix Programmer Assessment.	One (1) Copy for each student that successfully completes the ControlLogix Programmer Assessment.	Hard copy (paper) or electronic (PDF).	Primary and alternate TPOCs listed in this PWS.